EI 1110

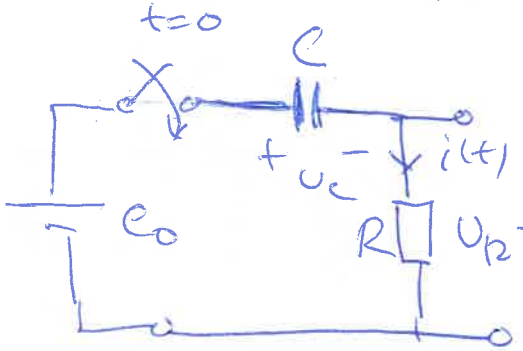
F 11

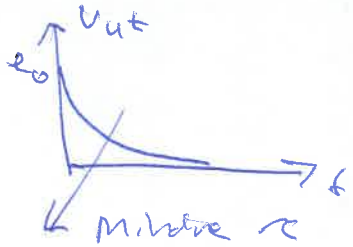
2025-01-14

Hans Edin

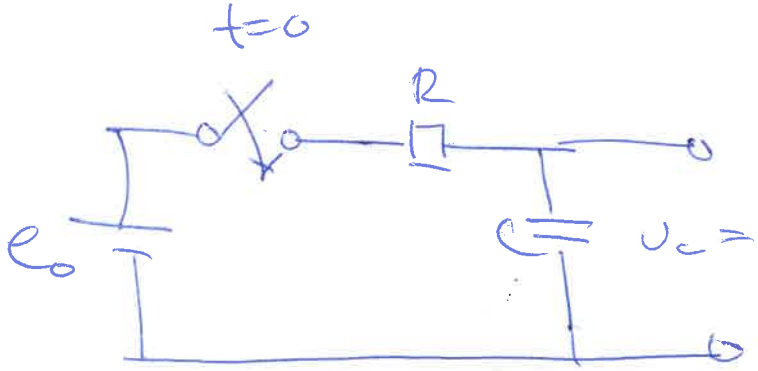
Enkla R-C och R-L kretsar:

Inkopplingsförlopp – transient uppförande:



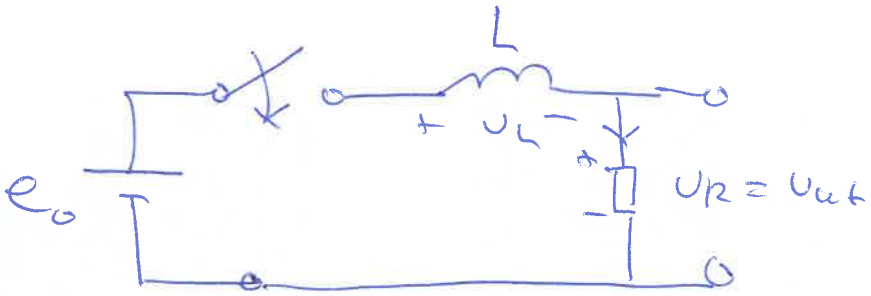


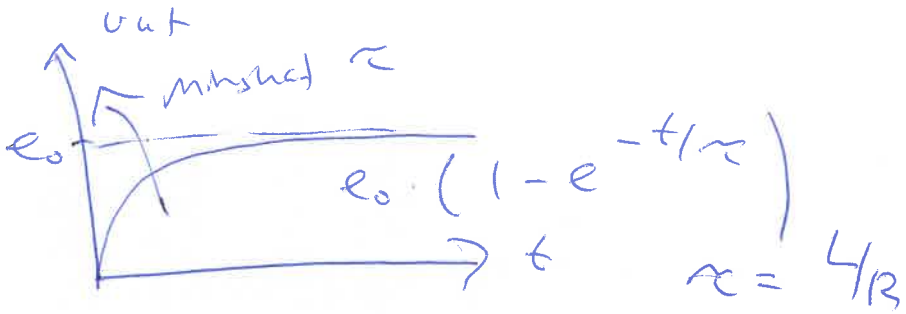
∴ Pulskrets – släpper fram en puls
– spärrar mot DC
– "Högpass" filter



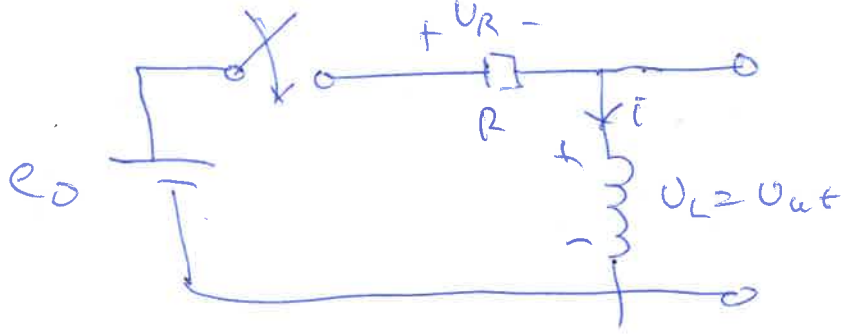


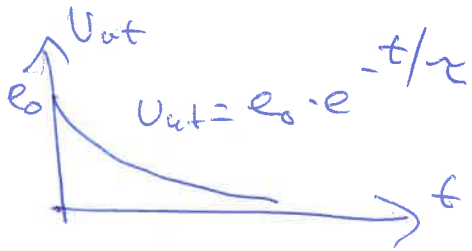
Släpper fram DC-spänning
Uppladdning av kondensator
Kortsluter höga frekvenser genom kondensator
– Lågpass filter





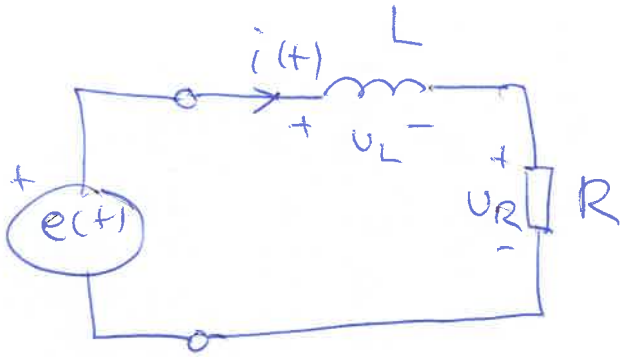
- Strömtång kvots
- Barkhållen snabba strömspikar (Blockerar höga frekvenser)
- Lågpass





- Stoppar från höga frekvenser
- Låga "frekvenser" DC kortsluts

R-L krets med $e_0 \cos(\omega t + \alpha)$-excitering:



$$U_L = L \cdot \frac{di(t)}{dt}$$

$$U_R = R \cdot i(t)$$

$$e(t) = e_0 \cdot \cos(\omega t - \alpha)$$

KVL: $e(t) = U_L + U_R$

$$\Rightarrow L \cdot \frac{di}{dt} + R\, i = e_0 \cdot \cos(\omega t - \alpha)$$

$$\Rightarrow \frac{di}{dt} + \frac{R}{L} \cdot i = \frac{e_0}{L} \cdot \cos(\omega t - \alpha)$$

$L/R = \tau$ tidskonstanten

$$\Rightarrow \frac{di}{dt} + \frac{1}{\tau} \cdot i = \frac{e_0}{L} \cdot \cos(\omega t - \alpha)$$

Använd metoden med integrerande faktorn: $e^{+t/\tau}$

$$\Rightarrow e^{t/\tau} \cdot \frac{di}{dt} + \frac{1}{\tau} e^{t/\tau} \cdot i = \frac{e_0}{L} \cdot e^{t/\tau} \cdot \cos(\omega t - \alpha)$$

$$\frac{d}{dt}\left(e^{t/\tau}\cdot i(t)\right) = \frac{e_0}{L}\cdot e^{t/\tau}\cdot\cos(\omega t-\alpha) \qquad \text{Integrera!}$$

$$\Rightarrow \quad i(t) = A\cdot e^{-t/\tau} + \frac{e_0}{L}\cdot e^{-t/\tau}\cdot \underbrace{\int_0^t e^{t'/\tau}\cos(\omega t'-\alpha)\,dt'}_{\text{(I)}}$$

$\boxed{i(0)=0}$ Begynnelsevärde

$$\text{(I)} = \int_0^t e^{t'/\tau}\cdot\cos(\underbrace{\omega t'-\alpha}_{x})\,dt' = \left\{\begin{array}{l} \omega t'-\alpha = x \\ dt' = \frac{1}{\omega}dx \\ 0\to -\alpha \\ t\to \omega t-\alpha \end{array}\right\} = \frac{1}{\omega}\int_{-\alpha}^{\omega t-\alpha} e^{\frac{x+\alpha}{\omega\tau}}\cdot\cos x\,dx =$$

$$= \frac{1}{\omega}e^{\alpha/\omega\tau}\cdot\int_{-\alpha}^{\omega t-\alpha} e^{\frac{1}{\omega\tau}\cdot x}\cdot\cos x\,dx = \left\{\begin{array}{l} \text{Tabell:} \\ \int e^{ax}\cos bx = \dfrac{e^{ax}\cdot(a\cos bx + b\sin bx)}{a^2+b^2} \end{array}\right\} =$$

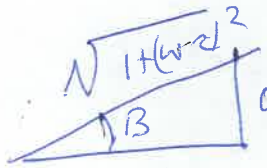
$$a=\frac{1}{\omega\tau}, \quad b=1$$

$$= \frac{1}{\omega}e^{\alpha/\omega\tau}\cdot\left[\frac{e^{x/\omega\tau}\cdot\left(\frac{1}{\omega\tau}\cdot\cos x + 1\cdot\sin x\right)}{\left(\frac{1}{\omega\tau}\right)^2+1}\right]_{-\alpha}^{\omega t-\alpha} =$$

$$= \frac{1}{\omega} e^{\alpha/\omega\tau} \left[\frac{e^{\frac{\omega t-\alpha}{\omega\tau}} \cdot \left(\frac{1}{\omega\tau} \cos(\omega t-\alpha) + \sin(\omega t-\alpha) \right) - e^{-\alpha/\omega\tau} \left(\frac{1}{\omega\tau} \cos(-\alpha) + \sin(-\alpha) \right)}{\left(\frac{1}{\omega\tau}\right)^2 + 1} \right]$$

$$= \frac{1}{\omega} \left[\frac{e^{t/\tau} \cdot \left(\frac{1}{\omega\tau} \cdot \cos(\omega t-\alpha) + \sin(\omega t-\alpha) \right) - \left(\frac{1}{\omega\tau} \cos\alpha - \sin\alpha \right)}{\left(\frac{1}{\omega\tau}\right)^2 + 1} \right] =$$

$$= \frac{1}{\omega} \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}} \left[\frac{e^{t/\tau} \cdot \left(\cos(\omega t-\alpha) + \omega\tau \cdot \sin(\omega t-\alpha) \right) - \left(\cos\alpha - \omega\tau \cdot \sin\alpha \right)}{\sqrt{1+(\omega\tau)^2}} \right] =$$



$$\Rightarrow \sin\beta = \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}} \qquad \cos\beta = \frac{1}{\sqrt{1+(\omega\tau)^2}}$$

$$= \frac{1}{\omega} \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}} \cdot \left[e^{t/\tau} \left(\underbrace{\cos\beta \cdot \cos(\omega t-\alpha) + \sin\beta \cdot \sin(\omega t-\alpha)}_{\cos(\omega t-\alpha-\beta)} \right) - \left(\underbrace{\cos\beta \cdot \cos\alpha - \sin\beta \cdot \sin\alpha}_{\cos(\alpha+\beta)} \right) \right]$$

$$=$$

$$= \frac{1}{\omega} \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}} \cdot \left[e^{-t/\tau} \cos(\omega t - (\alpha+\beta)) - \cos(\alpha+\beta) \right]$$

$$\Rightarrow \quad i(t) = A \cdot e^{-t/\tau} + e^{-t/\tau} \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}} \cdot \frac{e_0}{L} \left[e^{t/\tau} \cos(\omega t - (\alpha+\beta)) - \cos(\alpha+\beta) \right]$$

$$i(0) = 0 \Rightarrow \quad A = 0$$

$$\Rightarrow \quad i(t) = \underbrace{\frac{\tau}{\sqrt{1+(\omega\tau)^2}}}_{\text{Stationär del}} \cdot \frac{e_0}{L} \left[\underbrace{\cos(\omega t - (\alpha+\beta))}_{\text{Stationär del}} - \underbrace{e^{-t/\tau} \cdot \cos(\alpha+\beta)}_{\text{Avklingande del}} \right]$$

Stationär del

Avklingande del

$$\hat{I} = \frac{e_0}{L} \frac{\frac{L}{R}}{\sqrt{1+\left(\omega \frac{L}{R}\right)^2}} \cdot \frac{R}{R} =$$

$$= \frac{e_0}{\sqrt{R^2+(\omega L)^2}} \in Z \qquad Z = \sqrt{R^2+\omega L^2}$$

Totala impedansen.

$$\therefore \quad i(t) = \hat{I} \cdot \left[\cos\left(\omega t - (\alpha+\beta)\right) - e^{-t/\tau} \cdot \cos(\alpha+\beta) \right]$$

↑

Komplexa tal:

Observation: (Serieutveckling)

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \cdots$$

Alla termer ingår i $\cos x$ och $\sin x$

Försök: $e^x = \cos x \pm \sin x$

Nix, funkar inte

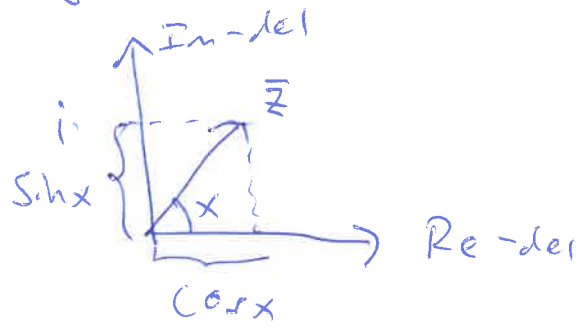
Ny algebra: Införa det komplexa "enheten" i ("operatorn") s.a.s. så att $\boxed{i^2 = -1}$

- Nu: $e^{ix} = 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \frac{(ix)^6}{6!} + \cdots$ (9)

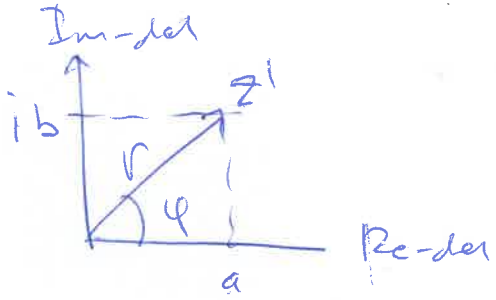
$$= \underbrace{1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}}_{\cos x} + i\cdot\underbrace{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)}_{\sin x}$$

$$\therefore\ e^{ix} = \cos x + i\sin x$$

Eftersom x är en vinkel så blir tolkningen av $e^{ix} = z$ en visare i ett plan:



Mera allmänt: φ - vinkeln, r - storleken/visarens längd



$$z = re^{i\varphi} = a + ib = \underbrace{r\cos\varphi}_{a} + i\underbrace{r\sin\varphi}_{b}$$

↑ polär form ↑ Rektangulär form

$$\varphi = \arctan\frac{b}{a}$$

$$r = \sqrt{a^2 + b^2}$$

Två komplexa tal: $\bar{z}_1 = r_1 e^{i\varphi_1} = a_1 + ib_1$

$\bar{z}_2 = r_2 e^{i\varphi_2} = a_2 + ib_2$

Addition/subtraktion: Använd rektangulär form

$$\bar{z}_1 \pm \bar{z}_2 = a_1 + ib_1 \pm a_2 + ib_2 = a_1 + a_2 \pm i(b_1 + b_2)$$

Multiplikation/division: Använd polär form:

$$\bar{z}_1 \cdot \bar{z}_2 = r_1 e^{i\varphi_1} \cdot r_2 e^{i\varphi_2} = r_1 r_2 e^{i(\varphi_1 + \varphi_2)}$$

$$= r_1 r_2 (\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2))$$

$$\frac{\bar{z}_1}{\bar{z}_2} = \frac{r_1}{r_2} \frac{e^{i\varphi_1}}{e^{i\varphi_2}} = \frac{r_1}{r_2} \cdot e^{i(\varphi_1 - \varphi_2)} =$$

$$\frac{r_1}{r_2} \cdot (\cos(\varphi_1 - \varphi_2) + i \sin(\varphi_1 - \varphi_2))$$

Konjugat: $\bar{z}^* = a - ib$

$$\bar{z} \cdot \bar{z}^* = (a + ib) \cdot (a - ib) = a^2 - aib + aib + b^2 = a^2 + b^2 = |\bar{z}|^2 = r^2$$